

GENITOURINARY PROCEDURES

- Prophylaxis of genitourinary procedures that enter the urinary tract should include coverage of likely pathogens as well as organisms from the most recent urologic source (urine culture, stone culture, etc.) within the preceding year, regardless of prior antimicrobial treatment. If the most recent positive urologic source culture was > 1 year ago, standard prophylaxis can be considered instead.
- All IV antibiotics should be given within an hour of surgery start time and oral antibiotics should be given 1-4 hours prior to surgery start time.
- General dosages are provided in the chart below, but these may need adjustment on a case-by-case basis for renal dysfunction, obesity, etc.

Nature of Operation	Likely pathogens	Recommended Regimens	Alternative Regimens
Cystography, urodynamic study, simple cystourethroscopy, shock wave lithotripsy Note: Prophylaxis only needed if risk factors present such as frailty, anatomic anomalies of urinary tract, chronic steroid use, and immunocompromise	Enteric gram-negative bacilli	Bactrim 1 DS PO OR Cephalexin 1g PO OR Amoxicillin/clavulanate 875/125 mg PO OR Cefazolin 2g IV/IM if <120 kg, 3g IV/IM if ≥120 kg	Gentamicin or Tobramycin 5 mg/kg (ideal body weight) or 120 mg IM OR Ciprofloxacin 500 mg PO (400 mg IV)
Cystourethroscopy with transurethral resection of bladder tumor and prostate Any biopsy or resection Fulguration Foreign body removal Urethral dilation Urethrotomy Submucosal injection (Botox)	Enteric gram-negative bacilli	Cefazolin 2g IV/IM if <120 kg, 3g IV/IM if ≥120 kg OR Amoxicillin/clavulanate 875/125 mg PO OR Bactrim 1 DS PO OR Cephalexin 1g PO	Gentamicin or Tobramycin 5 mg/kg (ideal body weight) IV or 120 mg IM OR Ciprofloxacin 500 mg PO (400 mg IV)

Nature of Operation	Likely pathogens	Recommended Regimens	Alternative Regimens
Ureteroscopy with stent placement and stone removal with asymptomatic bacteriuria	Enteric gram-negative bacilli	- Collect a urine culture 1-2 weeks prior to surgery - Provide 1-3 days of culture-directed oral antibiotics prior to scheduled procedure IF susceptible: Bactrim 1 DS PO BID* OR Cephalexin 1g PO TID* OR Amoxicillin/clavulanate 875/125 mg PO BID-TID* OR Nitrofurantoin 100 mg PO BID*	Ciprofloxacin 500-750 mg PO BID* Contact ID if no oral options available
Ureteroscopy with stent placement and stone removal with symptomatic UTI prior	Enteric gram-negative bacilli	Treat UTI as below, then a repeat urine culture post-treatment should be obtained to establish what bacteria to cover pre-operatively (note: <u>not</u> for test of cure) <u>UTI Treatment</u> IF culture susceptible: Bactrim 1 DS PO BID x 3 days* OR Cephalexin 1g PO TID x 5 days* OR Amoxicillin/clavulanate 875/125 mg PO BID-TID x 5 days* OR Nitrofurantoin 100 mg PO BID x 5 days* <u>Surgical Prophylaxis</u> Tailored IV antibiotic prophylaxis directly prior to procedure based on culture results, otherwise Cefazolin 2g IV/IM if <120 kg or 3g IV/IM if ≥120 kg	Ciprofloxacin 500-750 mg PO BID x 3 days* Contact ID if no oral options available

Nature of Operation	Likely pathogens	Recommended Regimens	Alternative Regimens
Percutaneous nephrolithotomy (PCNL) with high-risk features including positive urine culture, residual stone with prior positive stone culture, current indwelling stent or nephrostomy tube, severe hydronephrosis, continuous intermittent catheterization, renal transplant, severe immunocompromise, neurogenic bladder, urinary diversion, or chronic indwelling catheters (foley or SP tube)	Enteric gram-negative bacilli <i>Staphylococcus aureus</i> <i>Staphylococcus epidermidis</i>	<u>Adults without high-risk features:</u> No antibiotics in days preceding PCNL Cefazolin 2g IV/IM if <120 kg or 3g IV/IM if ≥120 kg Antibiotics should not be continued >24 hours unless there is concern for post-procedural sepsis <u>Adults with high-risk features:</u> Recommend early ID/ASP consultation in anticipation of PCNL Tailored oral antibiotics 3-5 days prior to PCNL, discuss with ID if no oral option available IF culture susceptible: Bactrim 1 DS PO BID x 3 days* OR Cephalexin 1g PO TID x 5 days* OR Amoxicillin/clavulanate 875/125 mg PO BID-TID x 5 days* OR Nitrofurantoin 100 mg PO BID x 5 days* Tailored IV antibiotic prophylaxis directly prior to procedure based on culture results, otherwise Cefazolin 2g IV/IM if <120 kg or 3g IV/IM if ≥120 kg Antibiotics only to continue >24 hours after procedure for sepsis / complicated UTI due to residual infected stone	Vancomycin 10-15 mg/kg IV x 1 AND Gentamicin or Tobramycin 5 mg/kg IV x 1 OR Unasyn 3g IV x 1 AND Gentamicin or Tobramycin 5 mg/kg IV x 1 Ciprofloxacin 500-750 mg PO BID x 3 days* Contact ID if no oral options are available

PROSTATE BIOPSY			
Nature of Operation	Likely pathogens	Recommended Regimens	Alternative Regimens
Prostate brachytherapy	<i>Staphylococcus aureus</i> <i>Staphylococcus epidermidis</i> <i>Streptococci spp.</i>	Cefazolin 2g IV/IM if <120 kg or 3g IV/IM if ≥120 kg	Vancomycin 10-15 mg/kg IV x 1
Transperineal prostate biopsy Routine use of antibiotic prophylaxis is not recommended but can be used in select patients deemed at high risk of infectious complications	None	Cefazolin 2g IV/IM if <120 kg or 3g IV/IM if ≥120 kg OR Cephalexin 1 gram PO 1 hour prior to procedure	Doxycycline 100 mg PO
Transrectal prostate biopsy	Enteric gram-negative bacilli	Best practice is to do a rectal swab prior to the procedure to culture bacteria and test for antimicrobial resistance. If no rectal swab obtained and patient has no recent history of Bactrim-resistant bacteria in urine nor recently received Bactrim: Bactrim (TMP-SMX) DS 1 tablet evening before biopsy and 1 tablet 1 hour prior to procedure	If patient has no recent history of ciprofloxacin-resistant bacteria in urine nor recently received ciprofloxacin: Ciprofloxacin 500 mg 1 tablet evening before biopsy and 1 tablet 1 hour prior to procedure OR If patient has no recent history of gentamicin-resistant bacteria in urine: Gentamicin 5 mg/kg IV x1 or 160 mg IM 1 hour prior to biopsy

OPEN, LAPAROSCOPIC, AND ROBOTIC SURGERY			
Nature of Operation	Likely pathogens	Recommended Regimens	Alternative Regimens
Adrenalectomy, Retroperitoneal/pelvic lymphadenectomy without entering the urinary tract	<i>S. aureus</i> <i>S. epidermidis</i> <i>Streptococci spp.</i>	Cefazolin 2g IV/IM if <120 kg or 3g IV/IM if ≥120 kg	Vancomycin 10-15 mg/kg IV x 1
Circumcision	Staphylococci	None unless diabetes mellitus or other risk factors; if risk factors present use: Cefazolin 2g IV/IM if <120 kg or 3g IV/IM if ≥120 kg	Vancomycin 10-15 mg/kg IV x 1
Genitourinary procedures involving small or large intestine Including urinary diversions, cystectomy with small bowel conduit, uretero-pelvic repair, colon conduits, radical cystectomy neobladder/ileostomy, etc	<i>S. aureus</i> <i>S. epidermidis</i> Streptococci Enteric gram-negative bacilli	Cefuroxime 500 mg IV/PO AND Metronidazole 500 mg IV/PO OR Cefoxitin 2g IV	Levofloxacin 500 mg IV/PO PLUS Metronidazole 500 mg IV/PO <i>If history of MRSA infection or colonization:</i> Vancomycin 10-15 mg/kg IV x 1 AND Gentamicin 5 mg/kg IV x 1
Penile prosthesis insertion, removal, & revision	<i>Staphylococci spp.</i> Enteric gram-negative bacilli	Cefazolin PLUS Gentamicin PLUS Fluconazole OR Vancomycin PLUS Gentamicin PLUS fluconazole Antibiotics should not be continued > 24 hours post-operatively unless there is concern for sepsis	
Testicular implants	<i>Staphylococci spp.</i> Enteric gram-negative bacilli	Cefazolin PLUS Gentamicin OR Unasyn PLUS Gentamicin OR Vancomycin PLUS Gentamicin	
Implanted prosthetic devices Artificial urinary sphincter and sacral nerve stimulators	Staphylococci Enteric gram-negative bacilli	Cefazolin PLUS Gentamicin OR Unasyn PLUS Gentamicin OR Vancomycin PLUS Gentamicin	

Nature of Operation	Likely pathogens	Recommended Regimens	Alternative Regimens
Vaginal urologic surgery Including urethral sling, fistulae repair, urethral diverticulectomy, urethropexy Open or laparoscopic procedure involving entry into the urinary tract Including urethroplasty; stricture repair including urethrectomy. Nephrectomy, partial or otherwise, urethrectomy, pyeloplasty, radical prostatectomy, partial cystectomy	Enteric gram-negative bacilli <i>S. aureus</i> <i>S. epidermidis</i> <i>Streptococci spp.</i> (vaginal surgery mostly)	Cefazolin 2g IV/IM if <120 kg or 3g IV/IM if ≥120 kg	Levofloxacin 500 mg IV/PO <i>If history of MRSA infection or colonization:</i> Vancomycin AND Gentamicin 5 mg/kg IV x 1
Vasectomy	<i>Staphylococci spp.</i> Enteric gram-negative bacilli	NOTE: Some experts do not recommend prophylaxis for all vasectomy cases. Prophylaxis may be considered based on complexity of procedure and comorbidities of patient. Cefazolin 2g IV/IM if <120 kg or 3g IV/IM if ≥120 kg	<i>If history of MRSA infection or colonization:</i> Vancomycin AND Gentamicin 5 mg/kg IV x 1
Inguinal and scrotal cases Including radical orchiectomy, reversals, varicocelectomy, hydrocelectomy	<i>Staphylococci spp.</i> Enteric gram-negative bacilli	Cefazolin 2g IV/IM if <120 kg or 3g IV/IM if ≥120 kg	<i>If history of MRSA infection or colonization:</i> Vancomycin AND Gentamicin 5 mg/kg IV x 1
Metoidioplasty	<i>S. aureus</i> <i>Streptococcus spp.</i>	Cefazolin 2g IV/IM if <120 kg or 3g IV/IM if ≥120 kg	Vancomycin 10-15 mg/kg IV x 1